OpportunityAI

AI-Powered Student Career Intelligence Platform

PRODUCT REQUIREMENTS DOCUMENT

Comprehensive Edition | Version 2.0

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VISION

To become the operating system for student careers by transforming fragmented opportunity discovery into a personalized AI-guided experience.

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01 Executive Summary

OpportunityAI is a next-generation, AI-powered student opportunity intelligence platform designed to solve one of the most critical yet overlooked problems in student career development: information overload and missed opportunities. By securely integrating with a student's email account, OpportunityAI automatically discovers, extracts, categorises, prioritises, and surfaces internships, hackathons, scholarships, research programs, workshops, and competitions — transforming a cluttered inbox into a powerful, personalised career action dashboard.

The platform leverages state-of-the-art large language models (LLMs), embedding-based matching, and rule-based intelligence to provide students with AI-generated summaries, deadline alerts, opportunity match scores, success probability predictions, and personalised career roadmaps. OpportunityAI is built for scale — starting with Indian university students and expanding to a global audience of early-career professionals.

Product	OpportunityAI — Student Career Intelligence Platform
Version	2.0 (Comprehensive PRD)
Target Launch	Q3 2026 (MVP)
Primary Market	Indian University Students (VIT Phase 1)
Tech Stack	Next.js · TypeScript · PostgreSQL · Prisma · Gemini/OpenAI · Vercel
Business Model	Freemium + Premium AI · University Partnerships · Recruiter Marketplace
5-Year Goal	Global AI Career Operating System for Students

02 Problem Statement & Opportunity

The Core Problem

Modern students are simultaneously over-communicated with and under-served by their digital tools. The average university student receives 50–150 emails per day from professors, departments, clubs, recruiters, and newsletters. Within this deluge lie career-changing opportunities — but identifying them requires time, attention, and organisation that most students simply cannot afford.

- **Information Overload:** Critical opportunities are buried in promotional and administrative email noise, making manual discovery impractical.
- **Deadline Blindness:** Internship, scholarship, and hackathon deadlines are scattered across emails with no centralised tracking.
- **Lack of Personalisation:** Existing job boards and opportunity portals require manual searching and do not adapt to individual student profiles.
- **No Priority Signal:** Students have no reliable way to assess which opportunities match their skills, year of study, or career goals.
- **Application Fragmentation:** Managing applications across dozens of platforms without a unified tracker leads to missed follow-ups and duplicated effort.
- **Inequitable Access:** Students without strong mentorship networks miss opportunities that well-connected peers discover via word-of-mouth.

Market Opportunity

Metric	Estimate	Source / Basis
Total university students globally	235 million+	UNESCO 2024
Indian university students	43 million+	AISHE 2023-24
Avg. career emails per student/week	15-40 emails	Founder research
Students missing at least 1 deadline/month	~68%	Primary survey (VIT)
Willingness to pay for premium career tools	~34%	Primary survey (VIT)
Addressable market (India, premium tier)	$120M+ annually	Estimated

03 Vision & Mission

Vision Statement

- To become the operating system for student careers — transforming fragmented opportunity discovery into a personalised, AI-guided, and outcome-oriented experience for every student, everywhere.

Mission Statement

- OpportunityAI empowers students to never miss a career-defining moment by intelligently discovering, organising, and prioritising every opportunity in their inbox — so they can focus on applying, not searching.

Core Values

Student-First	Every design and product decision prioritises student outcomes over platform metrics.
Privacy by Design	We request only what we need. Students retain full control of their data at all times.
AI with Integrity	Our AI recommendations are transparent, explainable, and free from undisclosed bias.
Inclusive Access	We build for the student without a mentor, not just the well-connected few.
Relentless Iteration	We ship fast, learn from users, and improve continuously.

04 Target Users & Personas

OpportunityAI targets university students across three geographic phases. The platform is designed to serve four core user personas, each with distinct goals, behaviours, and opportunity preferences.

Rollout Phases

Phase	Geography	Timeline	Target Reach
Phase 1	VIT Vellore & Chennai	2026 Q3–Q4	~25,000 students
Phase 2	Top 50 Indian Universities (IITs, NITs, Tier-1 privates)	2027 Q1–Q2	~2M students
Phase 3	Pan-India + South-East Asia	2027 Q3–Q4	~10M students
Phase 4	Global + Early-Career Professionals	2028+	235M+ students

User Personas

The Explorer (Year 1–2)

Primary Goal: Discover beginner-friendly programs, clubs, and workshops

Pain Point: Overwhelmed by college life; doesn't know where to start

Behaviour: Checks email infrequently; misses department announcements

Key Needs:

- Curated, beginner-friendly opportunity feed
- Simple onboarding with interest tagging
- Low-pressure reminders
- Explanatory AI summaries

The Competitor (Year 2–3)

Primary Goal: Maximise hackathon wins, internship offers, and competitive achievements

Pain Point: Scattered information across email, WhatsApp, and LinkedIn

Behaviour: Highly active; applies to 5–15 opportunities per semester

Key Needs:

- AI opportunity match scoring
- Team formation for hackathons
- Application tracker with status updates
- Competitive intelligence

The Researcher (Year 3–4 / Postgrad)

Primary Goal: Find research internships, lab positions, and academic conferences

Pain Point: Research opportunities are poorly advertised and hard to discover

Behaviour: Detail-oriented; values quality over quantity

Key Needs:

- Research-specific category filter
- Professor/lab recommendation engine
- Academic conference & grant discovery
- Publication opportunity alerts

The Placer (Final Year)

Primary Goal: Secure a full-time offer before graduation

Pain Point: Placement season is chaotic; managing multiple processes simultaneously

Behaviour: High urgency; tracks every application obsessively

Key Needs:

- Deadline urgency ranking
- Resume skill-gap analysis
- Mock interview resources
- Offer comparison dashboard
- Referral network insights

05 Core MVP Features

The MVP is scoped to deliver the minimum viable set of features that create real, measurable value for students from day one. All MVP features are required for launch.

Feature	Priority	Description
Google OAuth Login	MUST HAVE	Secure, one-click Google sign-in. No passwords stored. Read-only Gmail scope requested. Tokens encrypted at rest.
Gmail Read-Only Integration	MUST HAVE	Scheduled inbox scans via Gmail API. Fetches emails from the past 90 days on first login; incremental thereafter. Only opportunity-relevant emails are retained.
AI Opportunity Extraction Engine	MUST HAVE	LLM-powered pipeline classifies emails, extracts opportunity title, organisation, type, deadline, application link, eligibility, and prize/stipend information.
Deadline Detection & Parsing	MUST HAVE	Automatic detection of deadline dates from email body, subject, and attachments. Normalised to ISO 8601. Handles relative dates (e.g., 'by end of this month').
Opportunity Dashboard	MUST HAVE	Centralised, filterable, and sortable dashboard displaying all extracted opportunities. Cards show title, category, deadline, match score, and quick-action buttons.
Category Classification	MUST HAVE	Opportunities auto-classified into: Internship, Hackathon, Scholarship, Research, Workshop, Competition, Job, Event, Other.
Search & Filters	MUST HAVE	Full-text search + filters by category, deadline range, match score, status (bookmarked, applied, saved).
Bookmarks	MUST HAVE	Users can save opportunities. Bookmarks persist across sessions and are accessible from a dedicated tab.
Application Status Tracking	MUST HAVE	Users manually update application status: Interested → Applied → Under Review → Offer Received / Rejected.
Smart Reminders	MUST HAVE	Email and push notifications 7 days, 3 days, and 1 day before each tracked deadline. Urgency level adapts based on time remaining.
AI Opportunity Summaries	MUST HAVE	One-paragraph AI-generated summary for each opportunity including: what it is, who it's for, key benefits, and next steps.
User Profile & Interest Tagging	MUST HAVE	Students set their year, department, skills, and interest areas on onboarding. Profile drives recommendation personalisation.

06 Advanced AI Features

Post-MVP, OpportunityAI evolves from an extraction tool into a fully intelligent career co-pilot. The following advanced features are planned across V2–V5 releases, grouped by functional domain.

Intelligence & Matching

Feature	Version	Description
Opportunity Match Score	V2	Composite score (0–100) based on student profile, skills, year, past applications, and opportunity requirements. Displayed on every card.
Success Probability Prediction	V2	ML model trained on historical acceptance data to estimate likelihood of success. Factors in profile completeness, eligibility fit, and competition level.
Duplicate Detection	V2	Identifies semantically duplicate opportunities across emails and sources using embedding similarity. Merges into a single canonical record.
Opportunity Trend Forecasting	V3	Predicts which types of opportunities are emerging based on historical email frequency, industry trends, and seasonal patterns.

Career Development Tools

Feature	Version	Description
Resume Skill Gap Analysis	V2	Compares student's declared skills against requirements of top-matched opportunities. Generates a prioritised skill-building action list.
AI Career Roadmap Generator	V3	Produces a personalised 1–4 year career roadmap based on declared goals, current profile, and opportunity engagement history.
AI Application Advisor	V2	Provides tailored application tips, essay prompts, and preparation checklists for each specific opportunity.
AI Cover Letter Generator	V3	Drafts personalised cover letters using student profile data and opportunity details. Editable in-platform.
AI Resume Reviewer	V3	Analyses uploaded resume for ATS compatibility, skill alignment, and quality. Generates improvement recommendations.

Discovery & Aggregation

Feature	Version	Description
Scholarship Discovery Engine	V2	Dedicated ML pipeline trained to identify and extract scholarship-specific signals: GPA requirements, financial eligibility, essay topics.
Research Opportunity Finder	V2	Discovers professor openings, lab positions, and academic fellowships from departmental emails and external sources.

Feature	Version	Description
Public Opportunity Scraping	V4	Web crawlers aggregate opportunities from LinkedIn, Internshala, Devfolio, Unstop, and 50+ platforms. Combined with email data.
Browser Extension	V4	Captures opportunity details from any webpage the student visits. One-click 'Save to OpportunityAI' button.

Collaboration & Social

Feature	Version	Description
Team Formation for Hackathons	V3	Matches students seeking hackathon teammates based on skill sets, availability, and target competition type.
Recruiter Marketplace	V4	Verified recruiters post opportunities directly on OpportunityAI. Students can apply in-platform. Two-sided trust layer.
University Intelligence Layer	V3	University-specific opportunity feeds, department announcements, and placement cell integration per institution.

07 User Flows

Primary Onboarding Flow

Step	Stage	Description
1	**Landing Page**	Student visits OpportunityAI.com. Views value proposition, social proof, and CTA.
2	**Google Sign-In**	Clicks 'Connect with Google'. OAuth consent screen shows read-only Gmail scope. Student approves.
3	**Profile Setup**	Enters: Name, University, Year of Study, Department, Skills (tag-based), Career Interests (multi-select). Takes ~90 seconds.
4	**Initial Scan**	System triggers background job to scan past 90 days of emails. Progress indicator shown. First results appear in ~30–90 seconds.
5	**Dashboard Reveal**	Student sees their personalised opportunity dashboard. AI summaries and match scores displayed. Guided tooltip tour offered.
6	**First Action**	Student bookmarks first opportunity, sets a reminder, or updates application status. Activation complete.

Recurring Email Scan Flow

- Scheduler triggers scan every 4 hours (configurable) via background job
- Gmail API fetches new emails since last scan timestamp
- Classification model filters: Opportunity-relevant vs. Irrelevant
- Relevant emails pass through Extraction Pipeline (title, deadline, link, category, skills)
- Duplicate detection checks against existing opportunities for this user
- New opportunities persisted to database; user notified if high-priority (urgency score > 80)
- Dashboard updated in real-time via WebSocket or on next page load

08 Functional Requirements

Authentication & Authorisation

ID	Priority	Requirement
FR-01	CRITICAL	Users must authenticate exclusively via Google OAuth 2.0. No email/password authentication.
FR-02	CRITICAL	System must request only Gmail read-only scope (gmail.readonly). No write permissions.
FR-03	CRITICAL	OAuth tokens must be encrypted using AES-256 before database storage.
FR-04	HIGH	Users must be able to revoke Gmail access at any time from Settings. Revocation must delete all scanned email data within 24 hours.
FR-05	HIGH	Session tokens must expire after 24 hours of inactivity. Refresh token rotation required.

Email Scanning & Extraction

ID	Priority	Requirement
FR-06	CRITICAL	System must scan the past 90 days of Gmail on first login. Incremental scans thereafter (every 4 hours, configurable).
FR-07	CRITICAL	AI pipeline must extract: opportunity title, organisation name, opportunity type, application deadline, application link, eligibility criteria, and stipend/prize value (if present).
FR-08	CRITICAL	Extraction accuracy must meet or exceed 90% precision and 85% recall on the internal benchmark dataset.
FR-09	HIGH	System must handle multi-part MIME emails (HTML + plain text). Must parse deadline dates in at least 12 common formats.
FR-10	HIGH	Duplicate opportunities (same title + org within 30 days) must be detected and merged. User shown single canonical record.
FR-11	MEDIUM	System must process a full 90-day scan of up to 10,000 emails within 5 minutes.

Dashboard & Discovery

ID	Priority	Requirement
FR-12	CRITICAL	Dashboard must display all extracted opportunities in card format. Default sort: deadline ascending.
FR-13	CRITICAL	Each opportunity card must show: title, organisation, category badge, deadline, match score, and bookmark/apply CTAs.
FR-14	HIGH	Filters must include: category (multi-select), deadline range (date picker), match score range (slider), and application status.
FR-15	HIGH	Full-text search must return results within 500ms. Search indexed on title, organisation, and AI summary.
FR-16	MEDIUM	User must be able to manually add opportunities not found in email.

Notifications & Reminders

ID	Priority	Requirement
FR-17	CRITICAL	System must send email reminders at: 7 days, 3 days, and 24 hours before tracked deadlines.
FR-18	HIGH	Push notifications must be supported via browser (PWA) and mobile (future native app).
FR-19	HIGH	Users must be able to configure notification preferences: channels (email/push), frequency, and urgency threshold.
FR-20	MEDIUM	Daily opportunity digest email sent at user-preferred time (default: 8:00 AM local).

Application Tracking

ID	Priority	Requirement
FR-21	HIGH	Users must be able to track application status across 5 states: Interested, Applied, Under Review, Offer Received, Rejected.
FR-22	HIGH	Application status changes must be timestamped and displayed in a chronological activity log.
FR-23	MEDIUM	System must provide a Kanban-style view for managing applications across status columns.
FR-24	MEDIUM	Users must be able to attach notes, documents, and links to each tracked application.

AI Recommendations

ID	Priority	Requirement
FR-25	HIGH	System must compute and display a match score (0–100) for each opportunity based on student profile.
FR-26	HIGH	Recommendation engine must surface at least 5 personalised opportunity suggestions on dashboard load.
FR-27	MEDIUM	Users must be able to provide explicit feedback (thumbs up/down) on recommendations to improve future suggestions.
FR-28	MEDIUM	AI summaries must be regeneratable on demand and editable by the user.

09 Non-Functional Requirements

Performance	Dashboard load time < 2s (P95). API response < 500ms. Email scan < 5 min for 10k emails.
Availability	99.5% uptime SLA. Planned maintenance during 2–4 AM IST only.
Scalability	Horizontal scaling via Vercel edge. Database connection pooling. Support 100k concurrent users by Q4 2027.
Security	OWASP Top 10 compliance. Pen testing quarterly. All data encrypted in transit (TLS 1.3) and at rest (AES-256).
Accessibility	WCAG 2.1 AA compliance. Full keyboard navigation. Screen reader support. Min contrast ratio 4.5:1.
Data Retention	Email content not retained after extraction. Opportunity data retained for 12 months. User data deleted within 30 days of account deletion request.
Internationalisation	UTF-8 throughout. Date/time in user local timezone. Multi-language UI planned for V3 (Hindi, Tamil, Telugu priority).
Browser Support	Chrome 100+, Firefox 100+, Safari 15+, Edge 100+. Mobile: iOS Safari 15+, Chrome Android 100+.
API Rate Limits	Gmail API: Max 1B units/day (quota managed). Internal API: 1000 req/min per user. Burst: 50 req/sec.
Error Handling	All user-facing errors include actionable guidance. Error codes logged with trace IDs. P1 alerts < 5 min response.

10 AI & ML System Architecture

AI Pipeline Overview

The OpportunityAI extraction pipeline is a multi-stage LLM-powered system that transforms raw email content into structured, actionable opportunity records.

Stage 1: Ingestion
Raw email fetched from Gmail API. HTML stripped to clean text. Attachments flagged for separate processing.

Stage 2: Binary Classification
Fast classifier (fine-tuned BERT or LLM few-shot) determines: Is this email opportunity-relevant? Precision target: 95%+. Latency target: <100ms/email.

Stage 3: Structured Extraction
LLM (Gemini 1.5 Flash or GPT-4o-mini) extracts: title, org, category, deadline, link, eligibility, stipend, skills required. Output: JSON schema.

Stage 4: Deadline Normalisation
Parsed dates validated and normalised to UTC ISO 8601. Relative dates resolved against email received_at. Ambiguous dates flagged for user review.

Stage 5: Category Assignment
Multi-label classifier assigns one or more categories from: [Internship, Hackathon, Scholarship, Research, Workshop, Competition, Job, Event, Other].

Stage 6: Embedding Generation
Opportunity text embedded using text-embedding-3-small (OpenAI) or equivalent. Stored in pgvector for similarity search.

Stage 7: Match Score Computation
Cosine similarity between opportunity embedding and student profile embedding. Weighted by: skills match (40%), year/level (20%), category preference (20%), historical engagement (20%).

Stage 8: Recommendation Ranking
Top-K opportunities ranked by composite score. Collaborative filtering layer (V2+) adds popularity signal. Re-ranked list served to dashboard.

Model Selection & Rationale

Task	Model	Rationale	Latency Target
Binary Classification	Gemini Flash / GPT-4o-mini	Cost-effective; high throughput needed	<100ms
Structured Extraction	Gemini 1.5 Pro / GPT-4o	Complex JSON output; high accuracy required	<2s
Embedding Generation	text-embedding-3-small	Low cost; 1536 dimensions; pgvector compatible	<200ms
Summary Generation	Gemini Flash	Fluent, concise prose; cacheable per opportunity	<1s

Task	Model	Rationale	Latency Target
Career Roadmap	Gemini 1.5 Pro	Long-context; multi-step reasoning required	<5s
Cover Letter Gen	GPT-4o	Best writing quality; tone adaptation	<3s

11 System Architecture

Technology Stack

Layer	Technology	Justification
Frontend	Next.js 14 + TypeScript + Tailwind CSS	App Router, SSR/SSG, type safety, rapid UI dev
Backend	Next.js API Routes + Node.js	Unified codebase, serverless-ready, co-location
Database	PostgreSQL 16 + Prisma ORM	ACID compliance, pgvector for embeddings, type-safe queries
Vector Store	pgvector (PostgreSQL extension)	Co-located with primary DB; no additional infra
Authentication	Google OAuth 2.0 + NextAuth.js	Industry standard; zero password risk
AI / LLM	Google Gemini API + OpenAI API	Dual provider for redundancy and cost optimisation
Email Integration	Gmail API (Google Cloud)	Official API; read-only scope; robust quota
Background Jobs	Vercel Cron Jobs + BullMQ (Redis)	Scheduled scans; job queue for heavy processing
Caching	Redis (Upstash)	Session cache, rate limiting, job queue broker
File Storage	Vercel Blob / AWS S3	Resume uploads, attachment processing
Deployment	Vercel (Frontend + API) + Railway (workers)	Zero-config CI/CD; global edge CDN
Monitoring	Sentry + Vercel Analytics + PostHog	Error tracking, performance, product analytics
Notifications	Resend (transactional email) + Web Push	Reliable delivery; DMARC compliant

Architecture Diagram (Described)

- **Client Layer:** Next.js 14 PWA served from Vercel Edge CDN. Supports browser and future mobile clients.
- **API Layer:** Next.js API Routes handle all client requests. Rate-limited, authenticated, and request-validated.
- **Service Layer:** Modular services: EmailService, ExtractionService, RecommendationService, NotificationService, UserService.
- **AI Orchestration Layer:** LangChain-based orchestrator manages prompt routing, model fallbacks, and response parsing.
- **Data Layer:** PostgreSQL (primary data + pgvector). Redis (cache + queues). S3-compatible blob storage.
- **External APIs:** Gmail API, Google OAuth, Gemini API, OpenAI API, Resend Email API.
- **Background Workers:** Dedicated Node.js worker processes managed via BullMQ. Handles scans, extractions, and digest sends.

12 Database Schema

Table: users

Column	Type	Description
id	UUID	PRIMARY KEY, auto-generated
email	VARCHAR(255)	UNIQUE, NOT NULL — Google account email
name	VARCHAR(255)	Full name from Google profile
google_id	VARCHAR(255)	UNIQUE — Google sub claim
university	VARCHAR(255)	University name (user-entered or inferred)
year_of_study	INTEGER	1–6 (or NULL for postgrad)
department	VARCHAR(255)	Branch / department
skills	TEXT[]	Array of skill tags
interests	TEXT[]	Array of interest/category preferences
profile_embedding	VECTOR(1536)	pgvector — for similarity matching
gmail_refresh_token	TEXT	Encrypted AES-256; nullable (revocation sets NULL)
last_scan_at	TIMESTAMPTZ	Timestamp of most recent Gmail scan
created_at	TIMESTAMPTZ	Account creation timestamp
deleted_at	TIMESTAMPTZ	Soft delete; triggers data purge job

Table: opportunities

Column	Type	Description
id	UUID	PRIMARY KEY
user_id	UUID	FK → users.id (owner)
title	VARCHAR(500)	Extracted opportunity title
organisation	VARCHAR(255)	Extracted organisation/company
category	opportunity_category	ENUM: internship, hackathon, scholarship, research, etc.
deadline	DATE	Normalised application deadline (UTC)
deadline_confidence	FLOAT	0.0–1.0 confidence of deadline extraction
application_link	TEXT	Primary application URL
summary	TEXT	AI-generated opportunity summary
raw_email_id	UUID	FK → emails.id (source)
match_score	INTEGER	0–100 relevance score for this user
skills_required	TEXT[]	Extracted required skills
stipend_value	VARCHAR(100)	Prize/stipend text if found
eligibility_text	TEXT	Raw eligibility criteria from email
embedding	VECTOR(1536)	pgvector — for similarity search
is_duplicate	BOOLEAN	TRUE if merged into another record
canonical_id	UUID	FK → opportunities.id (master record)
created_at	TIMESTAMPTZ	Extraction timestamp

Table: applications

Column	Type	Description
id	UUID	PRIMARY KEY
user_id	UUID	FK → users.id
opportunity_id	UUID	FK → opportunities.id
status	app_status	ENUM: interested, applied, under_review, offer_received, rejected
applied_at	TIMESTAMPTZ	NULL until status = applied
notes	TEXT	User free-text notes
updated_at	TIMESTAMPTZ	Last status change timestamp

Table: notifications

Column	Type	Description
id	UUID	PRIMARY KEY
user_id	UUID	FK → users.id
opportunity_id	UUID	FK → opportunities.id (nullable)
type	notification_type	ENUM: deadline_reminder, new_opportunity, digest, system
channel	VARCHAR(20)	email | push | in_app
message	TEXT	Notification body
sent_at	TIMESTAMPTZ	Delivery timestamp
read_at	TIMESTAMPTZ	NULL until opened
status	VARCHAR(20)	pending | sent | failed | read

13 Security & Privacy

Security Principles

- Least Privilege: Request only gmail.readonly scope. Never request write or send permissions.
- Zero Password Storage: Authentication exclusively via Google OAuth. No credentials stored.
- Data Minimisation: Only opportunity-relevant data persisted. Raw email content discarded post-extraction.
- User Sovereignty: Users can revoke access and delete all data at any time. Processed within 30 days.
- Transparency: Clear privacy policy. Audit log available to users on request.

Control	Implementation	Standard
OAuth Token Storage	AES-256 encryption; tokens stored in encrypted column; keys in environment secrets	OWASP A02
Transport Security	TLS 1.3 enforced on all connections; HSTS preloading; no HTTP fallback	OWASP A02
SQL Injection Prevention	Prisma ORM with parameterised queries; no raw SQL from user input	OWASP A03
XSS Prevention	Next.js built-in CSP; React's DOM escaping; strict Content-Security-Policy header	OWASP A03
CSRF Protection	SameSite=Strict cookies; CSRF tokens on state-changing API endpoints	OWASP A01
Rate Limiting	1000 req/min per user; 50 req/sec burst; Redis-backed; 429 responses with retry headers	OWASP A04
Authentication Verification	JWT validation on every API route; token expiry 24h; refresh rotation	OWASP A07
Role-Based Access Control	Roles: Student, Admin, University Partner, Recruiter. Row-level security via Prisma middleware.	OWASP A01
Audit Logging	All data access, modifications, and deletions logged with user ID, IP, timestamp, and action	SOC 2 Type II
Penetration Testing	Quarterly automated scanning (OWASP ZAP); annual third-party pen test	ISO 27001
GDPR / DPDP Compliance	Data Processing Agreement available; consent captured at onboarding; erasure within 30 days	GDPR Art. 17
Secret Management	All API keys and secrets in Vercel Environment Variables; never in codebase or logs	OWASP A02

14 Recommendation Engine

The recommendation engine evolves through three maturity phases, each adding progressively more sophisticated signals:

Phase 1 — Rule-Based (MVP)

- Profile-keyword matching: student skills ∩ opportunity skills_required
- Category preference weighting from onboarding interest selections
- Deadline urgency boost: opportunities expiring within 7 days ranked higher
- Eligibility filter: year_of_study and department hard constraints

Phase 2 — Embedding-Based (V2)

- Student profile vector generated from: skills, interests, past applications, browsing history
- Opportunity vector generated from: title, summary, skills_required, organisation
- Cosine similarity score (0–1.0) as primary ranking signal
- Weighted composite: similarity (40%) + deadline urgency (20%) + category pref (20%) + recency (20%)

Phase 3 — Collaborative Filtering (V3)

- User-opportunity interaction matrix built from bookmark, apply, and view events
- Implicit feedback: dwell time on opportunity cards, click-through on AI summaries
- Matrix factorisation to surface opportunities popular with similar student profiles
- Popularity signal: opportunities with high engagement from same university/department

15 Notification Engine

Notification Type	Trigger	Channel	Timing
Deadline Reminder (High)	Deadline ≤ 24 hours; application status = Interested or Applied	Email + Push	24h before
Deadline Reminder (Medium)	Deadline ≤ 3 days; opportunity bookmarked	Email + Push	3 days before
Deadline Reminder (Low)	Deadline ≤ 7 days; opportunity viewed	Email	7 days before
New Opportunity Alert	High match score (≥85) opportunity extracted	Push + In-App	Real-time
Daily Digest	Configurable; summarises new + expiring opportunities	Email	User-set time (default 8AM)
Weekly Roundup	Top 5 opportunities of the week + career tip	Email	Every Monday
Application Milestone	Status changed to Offer Received	Email + In-App	On event
Scan Complete	Initial 90-day scan finishes	In-App	On completion
Smart Urgency Alert	Urgency score crosses threshold (configurable)	Push	Real-time

16 Analytics Dashboard

Student-Facing Analytics

- Opportunities discovered (total + by category) — monthly trend chart
- Deadlines met vs missed — success rate percentage
- Applications submitted — funnel: Interested → Applied → Offer
- Match score distribution — histogram of scored opportunities
- Top categories for this student — donut chart
- Upcoming deadlines calendar — 30-day forward view
- Skill gap report — top 5 missing skills across bookmarked opportunities

Admin / University Partner Analytics

- Daily/Monthly Active Users (DAU/MAU) by university
- Opportunities extracted per day — volume and quality trend
- Email scan success rate — extraction accuracy tracking
- Notification engagement rates — open rate, click-through rate
- Recommendation click-through rate (CTR) — by category and score band
- Top opportunities by engagement — most bookmarked / most applied
- Cohort retention analysis — D1, D7, D30, D90
- Placement outcomes (opt-in) — reported offer conversions

17 Product Roadmap

Q3 2026 — MVP Launch [ACTIVE]

- Google OAuth + Gmail integration
- AI extraction pipeline (Gemini Flash)
- Opportunity dashboard with filters
- Deadline detection and reminders
- Bookmarking and application status tracking
- Basic user profile + interest tagging
- VIT Vellore beta launch (500 users)

Q4 2026 — V2 — Intelligence [PLANNED]

- Embedding-based match scoring
- AI opportunity summaries
- Scholarship & research-specific pipelines
- Resume skill gap analysis
- Kanban application tracker
- Daily digest email
- Analytics dashboard (student-facing)
- Expand to VIT Chennai + 5 partner universities

Q1–Q2 2027 — V3 — Scale [PLANNED]

- Multi-university rollout (50 universities)
- Microsoft Outlook / Office 365 integration
- AI career roadmap generator
- AI cover letter generator
- Hackathon team formation feature
- University intelligence layer (per-institution feeds)
- Mobile PWA enhancements

Q3–Q4 2027 — V4 — Aggregation [FUTURE]

- Public opportunity scraping (LinkedIn, Internshala, Devfolio, Unstop)
- Browser extension (Chrome + Firefox)
- Recruiter marketplace (beta)
- Mobile native apps (iOS + Android)
- Enterprise analytics portal for universities
- Embedding-based collaborative filtering

2028+ — V5 — Global Career OS [FUTURE]

- Global opportunity marketplace
- AI-powered application auto-fill (with consent)
- Interview preparation AI coach
- Salary benchmarking and offer evaluation
- Alumni mentorship network integration
- Early-career professional expansion (0–3 YOE)
- B2B API for university career offices

18 Business Model

Revenue Streams

Stream	Description	Est. Launch	Target ARR (Year 2)
Freemium → Premium (Students)	Free: 50 opps/month, basic filters. Premium (■199/mo): unlimited extractions, AI insights, career roadmap, priority support.	Q3 2026	■2.5 Cr
University Partnerships	Annual SaaS license per institution. Includes custom dashboard, student analytics, placement tracking, and API access.	Q1 2027	■3 Cr
Recruiter / Company Partnerships	Companies pay to post verified opportunities directly on OpportunityAI. Performance-based pricing (per-application or per-placement).	Q3 2027	■1.5 Cr
Sponsored Opportunities	Verified sponsors pay for featured placement on the dashboard. Clearly labelled as sponsored. CPM / CPC model.	Q2 2027	■0.8 Cr
Premium AI Add-ons	Standalone purchases: Resume Review (■99), Cover Letter Pack (■149), Mock Interview Prep (■299).	Q1 2027	■0.7 Cr
Enterprise Analytics API	Universities and government bodies access anonymised trend data and placement intelligence via API.	2028	■1.5 Cr

Pricing Tiers

Feature	Free	Premium (■199/mo)	University (Custom)
Opportunities / month	50	Unlimited	Unlimited
AI Summaries	5/month	Unlimited	Unlimited
Match Scoring	Basic	Advanced (weighted)	Advanced + Custom weights
Deadline Reminders	Email only	Email + Push	Email + Push + SMS
Application Tracker	Up to 10	Unlimited	Unlimited
Career Roadmap	—	✓	✓
Resume Skill Gap	—	✓	✓
Cover Letter AI	—	1/month	Unlimited
Analytics Dashboard	Basic	Full	Full + Institution view

Feature	Free	Premium (■199/mo)	University (Custom)
Support	Community	Priority email	Dedicated CSM
Data Export	—	CSV	CSV + API

19 Success Metrics & KPIs

Acquisition

KPI	Target (MVP)	Target (6 Months)	Target (12 Months)
New Signups per Week	100 (Q3 2026)	1,000 (Q4 2026)	10,000 (Q2 2027)
Organic vs Paid Signup Ratio	> 70% organic	—	—
Referral Rate	—	> 15% of signups	> 20% of signups

Activation

KPI	Target (MVP)	Target (6 Months)	Target (12 Months)
Gmail Connected Rate	> 85% of signups	> 90%	> 92%
First Opportunity Viewed	> 90% within first session	> 92%	> 95%
Profile Completion Rate	> 70% complete onboarding	> 80%	> 85%

Engagement

KPI	Target (MVP)	Target (6 Months)	Target (12 Months)
Daily Active Users (DAU)	200 (Month 1)	2,000 (Month 3)	20,000 (Month 12)
Avg. Sessions per User / Week	> 2.5	> 3.0	> 3.5
Opportunities Bookmarked per DAU	> 1.5/session	> 2.0	> 2.5
Reminder Engagement Rate	> 35% open rate	> 40%	> 45%

Retention

KPI	Target (MVP)	Target (6 Months)	Target (12 Months)
D7 Retention	> 45%	> 50%	> 55%
D30 Retention	> 25%	> 30%	> 35%
Monthly Churn (Premium)	< 8%	< 6%	< 4%

Revenue

KPI	Target (MVP)	Target (6 Months)	Target (12 Months)
Free-to-Premium Conversion	> 3% by M3	> 5% by M6	> 8% by M12

KPI	Target (MVP)	Target (6 Months)	Target (12 Months)
Monthly Recurring Revenue (MRR)	■50K (Q4 2026)	■2L (Q2 2027)	■20L (Q4 2027)
Revenue per University Partner	■3L ARR	■5L ARR	■10L ARR

Product Quality

KPI	Target (MVP)	Target (6 Months)	Target (12 Months)
Extraction Precision	> 90%	> 93%	> 95%
Extraction Recall	> 85%	> 88%	> 90%
Recommendation CTR	> 12%	> 18%	> 25%
NPS Score	> 40	> 50	> 60

20 Risk Register & Mitigations

ID	Risk	Likelihood	Impact	Severity
R-01	User Trust & Privacy Concerns	**HIGH**	**HIGH**	**CRITICAL**
R-02	Gmail API Quota Exhaustion	**MEDIUM**	**HIGH**	**HIGH**
R-03	Low Extraction Accuracy	**MEDIUM**	**HIGH**	**HIGH**
R-04	User Acquisition Cost Escalation	**MEDIUM**	**MEDIUM**	**MEDIUM**
R-05	Competitive Displacement	**MEDIUM**	**MEDIUM**	**MEDIUM**
R-06	Regulatory / Compliance Risk	**LOW**	**HIGH**	**HIGH**
R-07	AI Hallucination in Summaries	**HIGH**	**MEDIUM**	**MEDIUM**
R-08	Infrastructure Cost Overrun	**MEDIUM**	**MEDIUM**	**MEDIUM**

Mitigation Strategies

R-01 — User Trust & Privacy Concerns:
Risk: Users reluctant to grant Gmail access; privacy fears.

Mitigation: Clear, plain-language privacy policy. Gmail read-only scope only. On-device data processing where feasible. Third-party privacy audit pre-launch. 'No email storage' guarantee prominently displayed.

R-02 — Gmail API Quota Exhaustion:
Risk: Gmail API rate limits exceeded at scale, causing scan failures.

Mitigation: Tiered scan scheduling (premium users scanned first). Exponential backoff and retry logic. Quota monitoring with auto-alerts at 70% usage. Request increase from Google Cloud as scale grows.

R-03 — Low Extraction Accuracy:
Risk: AI fails to correctly extract deadlines or links, eroding user trust.

Mitigation: Continuous benchmark evaluation against labelled dataset (500+ emails). Human-in-the-loop validation for low-confidence extractions. A/B testing of prompts monthly. User feedback ('Report incorrect extraction') integrated into training pipeline.

R-04 — User Acquisition Cost Escalation:
Risk: CAC exceeds LTV if organic growth stalls.

Mitigation: University ambassador program (peer-to-peer referrals). Free university partnership model for initial institutions. Content marketing (SEO + LinkedIn). Viral loop: share opportunity with peers.

R-05 — Competitive Displacement:
Risk: LinkedIn, Internshala, or a well-funded startup copies core functionality.

Mitigation: First-mover advantage in Indian university email intelligence. Deep Gmail integration is a technical moat. University partnership lock-in. Network effects from team formation feature.

R-06 — Regulatory / Compliance Risk:
Risk: DPDP Act 2023 (India) or GDPR requirements create compliance overhead or fines.

Mitigation: Legal counsel engaged pre-launch. Privacy-by-design architecture. Data residency in India (AWS Mumbai region). DPA available for institutional customers. Regular compliance review cycle.

R-07 — AI Hallucination in Summaries:
Risk: LLM generates incorrect or misleading opportunity summaries.

Mitigation: Source attribution: every summary includes link to original email. Grounding: prompts explicitly reference extracted facts. User can flag 'Inaccurate Summary'. Temperature set to 0.2 for extraction tasks.

R-08 — Infrastructure Cost Overrun:
Risk: LLM API costs or compute costs scale faster than revenue.

Mitigation: Cost-per-extraction tracked in real-time. Cheaper models (Flash/mini) for high-volume tasks. Caching of LLM responses (same email = same extraction). Freemium limits align with cost structure.

21 Future Vision (5-Year)

OpportunityAI's long-term ambition extends far beyond email parsing. By 2030, we envision a world where every student — regardless of institution prestige, mentor network, or socioeconomic background — has access to the same quality of career intelligence as the most well-connected students at elite universities.

2026	**Foundation** MVP live at VIT. Core extraction and reminders working. 1,000 students onboarded. Product-market fit validated.
2027	**Scale** 50+ Indian universities. Recommendation engine v2. Outlook integration. ■2Cr ARR. University partnership model proven.
2028	**Aggregation** Public opportunity scraping live. Browser extension. Mobile apps. Recruiter marketplace. Global beta (UK, US, SEA). ■10Cr ARR.
2029	**Intelligence** AI application auto-fill. Interview prep coach. Placement outcome tracking (opt-in). 1M+ users globally. Series A funding.
2030	**Operating System** Global career marketplace. AI-powered mentorship matching. Employer hiring intelligence. B2B enterprise API. 10M+ students. Category-defining platform.

22 Appendix

A. Opportunity Category Definitions

Category	Definition	Examples
Internship	Paid or unpaid work experience at a company, typically 1–6 months	SDE Intern (Google), Marketing Intern (Startup)
Hackathon	Time-bounded competitive coding or innovation event	Smart India Hackathon, HackMIT
Scholarship	Financial award based on merit, need, or identity criteria	Tata Scholarship, Google Generation Scholarship
Research	Academic or industry research position, fellowship, or conference	IISER Summer Research, DAAD Fellowship
Workshop	Skill-building session, typically 1–5 days	AWS ML Workshop, Design Thinking Bootcamp
Competition	Non-hackathon competitive event with prizes or recognition	Case competition, Robocon, Quant Challenge
Job	Full-time or part-time employment opportunity	SDE-1 (Amazon), Analyst (McKinsey)
Event	Conference, summit, or networking event	TechFest IIT Bombay, Nasscom Product Conclave
Other	Opportunities that don't fit above categories	Open source contribution programs, volunteer work

B. Glossary

LLM	Large Language Model — AI model trained on large text corpora to understand and generate natural language (e.g., Gemini, GPT-4).
Embedding	A numerical vector representation of text that captures semantic meaning. Similar texts have similar embeddings.
pgvector	An open-source PostgreSQL extension for storing and querying vector embeddings natively.
OAuth 2.0	An authorisation framework enabling third-party applications to access user data without exposing credentials.
Match Score	A composite 0–100 relevance score indicating how well an opportunity fits a specific student's profile.
Extraction Pipeline	The multi-stage AI process that converts raw email text into structured opportunity records.
Cosine Similarity	A mathematical measure of similarity between two vectors, ranging from -1 (opposite) to 1 (identical).
BullMQ	A Node.js library for managing Redis-backed background job queues with priority, retries, and scheduling.

PWA	Progressive Web App — a web application that behaves like a native mobile app with offline support and push notifications.
DPDP	Digital Personal Data Protection Act (India, 2023) — India's primary data protection legislation.
DAU/MAU	Daily Active Users / Monthly Active Users — standard engagement metrics for digital products.
ATS	Applicant Tracking System — software used by employers to manage job applications.

C. Related Documents

- OpportunityAI Technical Architecture Spec (v1.0)
- OpportunityAI API Documentation
- OpportunityAI Privacy Policy (Draft)
- OpportunityAI User Research Report — VIT Beta
- OpportunityAI Competitor Analysis
- OpportunityAI Go-to-Market Strategy
- OpportunityAI Brand Guidelines
